Lamiss Ben Haj Saad

Software Engineer

+216 52 127 818 | My Portfolio | lamis.benhajsaad@issatso.u-sousse.tn
/in/lamis-ben-haj-saad | github.com/Lamis-haj-saad

Curious and adaptable Software Engineering graduate with experience in AI, data science, and multiple programming languages. Passionate about building data-driven solutions and continuously exploring emerging technologies.

Education

Software Engineering – Software Architecture *Sep 2022 – Jul 2025*
Higher Institute of Applied Sciences and Technology of Sousse

- **Relevant Coursework:** Software Development Lifecycle (SDLC), Software Architecture and Design, Artificial Intelligence, Machine Learning, Computer Vision, Data Structures & Algorithms, Database Systems, Data Science, Cloud Computing, DevOps, Mobile Development, Web Technologies, Programming Languages and Frameworks.

Pre-engineering: Maths, Physics, Informatics *Sep 2020 – Jun 2022*
Higher Institute of Applied Sciences and Technology of Sousse

- **Relevant Coursework:** Data Structures and Algorithms, Prob & Stats, Linear Algebra, Analysis, Networking, System Administration, System Design, Object Oriented Programming.

Internships & Experience

Full Stack Developer – Internship *Feb 2025 (4 Months)*
DECADE

Project: Design, development and integration of an AI-based assistant into a catering services platform.

- Completed an accelerated training in ReactJS, Magento, Elasticsearch and E-commerce.
- Analyzed functional and technical requirements to define specifications for an intelligent assistant.
- Developed the client interface using **ReactJS** and built a **Node.js** server to process API requests.
- Implemented speech transcription using Speech-to-Text models to process voice input.
- Integrated NLP modules to interpret user intentions and generate personalized recommendations.
- Applied **prompt engineering** to optimize model responses and improve recommendation accuracy.
- Researched and documented the internal architecture of conversational agents (Transformer models, NLU/NLG).

Methodologies:

- **Scrum** for iterative development and sprint planning.
- **Layered Architecture:** The Node.js server executes intelligent models (NLP, Transcription), while the interface uses Elasticsearch for direct data access.
- **Prompt Engineering** to optimize model responses and improve the accuracy of recommendations.

Stack:
Front-end: ReactJS, TypeScript, SCSS
Back-end: Node.js, JavaScript, Express.js, REST API
Data & Communication: JSON, Axios
Search Engine: Elasticsearch
AI Models: vosk-model-fr-0.22, OpenHermes-2.5-Mistral-7B-GGUF

Hangfire Service Developer – Internship *Aug 2024 (1 Month)*
TWINKLE-IT

Project: Design and development of an automated task scheduling service.

- Designed and developped a background job scheduling service, enabling users to create and schedule tasks.
- Integrated **Hangfire** to manage background tasks within the service, eliminating the need for manual task triggering.
- Transformed the service into an API enabling backgroung task management.

Stack: C#, ASP.NET Core, Hangfire, Entity Framework, PostgreSQL

Front-End Developer – Internship *Aug 2022 (1 Month)*
E-DIGIAPP

Project: Development of a showcase website for a metal curtain production company.

- Translated UI designs into responsive, dynamic web pages.

Stack: HTML, CSS, jQuery, JavaScript

Projects

ISSATSO+ – Web Platform for ISSAT Sousse *2 Months*
Team: 3 People

- Built discussion forums enabling communication between students and teachers.
- Digitized administrative procedures including registration and document requests.
- Developed a quiz and test module allowing teachers to create assessments for student evaluation.
- Automated attendance management through a **signature recognition system** using digital fingerprints.

Methodologies: Waterfall methodology, MVC (Model-View-Controller) architecture.
Stack: HTML, CSS, JavaScript, jQuery, Bootstrap, PHP 8.3

Traffic Prediction – Data Science & Machine Learning *2 Weeks*

- Cleaned and prepared traffic data : engineered time and speed features to improve model performance.
- Conducted exploratory data analysis to visualize traffic patterns by hour and day.
- Built and evaluated a **Random Forest Regressor** using MAE and RMSE metrics.

Stack: Python, Pandas, NumPy, Scikit-learn, Matplotlib

Digit Recognition System – Computer Vision *2 Weeks*

- Built an image processing pipeline (grayscale, Gaussian filtering, Canny edge detection) for digit extraction.
- Trained a **KNN** classifier on 5,000 digit samples; overlaid predictions directly on source images.

Stack: Python, OpenCV, NumPy

Distributed Banking Management System *1 Month*
Team: 2 People | Methodology: Waterfall

- Implemented data distribution across multiple sites.
- Enforced secure access control to protect sensitive banking data.
- Managed real-time data updates and automated transactions.

Stack: Advanced SQL, functions, cursors, relational-object databases, stored procedures, triggers

---

WhatsApp Clone – Mobile Application *1 Month*

- Implemented user profile creation, viewing, and updating.
- Built private and group instant messaging with real-time status updates (online/offline).
- Enabled sending of text messages and image files.

Stack: JavaScript, React Native, CSS

Connect Four Game *2 Weeks*

- Implemented game logic, player turn management, score tracking, and real-time updates.

Stack: Python, Pygame, Tkinter

Skills

Languages	English (fluent), French (fluent), Arabic (native)
Programming	JavaScript, TypeScript, ReactJS, React Native, Node.js, Express.js, PHP, Symfony, Spring Boot, C#, .NET, XAF Blazor, Java (OOP, JavaFX), Android (SQLite), C/C++, Python, MATLAB
Databases	MySQL, PostgreSQL, PL/SQL, Oracle, Elasticsearch, Data Modeling, Data Warehousing
AI & ML	Machine Learning, Computer Vision, Prompt Engineering, Scikit-learn, Pandas, NumPy
Cloud & DevOps	AWS (Fundamentals), Git, GitHub/GitLab, Linux, DevOps Fundamentals
Methodologies	Agile/Scrum, UML, Software Architecture, REST APIs, Web 2.0 & Web 3.0
Soft Skills	Problem Solving, Social Intelligence, Effective Communication, Team Collaboration, Adaptability, Strong Memory, Attention to Details

Volunteering

Cyber Trace ISSATSo *(Financial Team Member)* *Sep 2021 – May 2022*
Google Developpers Students Club *(Member)* *Sep 2021 – Jul 2025*
ARSII *(Member)* *Sep 2023 – Jul 2024*

Certificates

- **Building Database Applications in PHP** – University of Michigan *Apr 2024*
- **Agile Project Management** – Google *Jun 2024*
- **Python** – Kaggle *Aug 2024*
- **Pandas** – Kaggle *Aug 2024*
- **Introduction to Machine Learning** – Kaggle *Sep 2024*